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#88 RICE 2017 TECHNOLOGY BULLETIN

PHILIPPINE RICE RESEARCH INSTITUTE

PALAYCHECK SYSTEM FOR UPLAND RICE FARMING

SECOND EDITION



Rice Technology Bulletin Series

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FOREWORD

Upland rice farming is a community-based practice that relies mainly on local knowledge, particularly on the production aspects of traditional rice varieties. The development of the upland rice ecosystem offers an opportunity to augment the country's rice supply and ensure food sufficiency of the upland farming communities. Hence, the Upland Rice Development Program (URDP) was crafted.

URDP targets a yield of 2-3 t/ha from the usual 1- 2 t/ha. Also, the program aims to accelerate the adoption and integration of PalayCheck Upland Rice Farming technologies into the farming systems of participating communities.

PalayCheck is an integrated crop management system for rice, which provides recommendations on what to do and what to achieve based on best management practices for a particular agro-ecological condition. It seeks to integrate these recommendations at the farm-level, taking into account the interactions among management practices & other factors affecting yield, grain quality, and environment.

It is hoped that through this bulletin, the income of upland farmers could be increased and self-sufficient food communities would be established.

INTRODUCTION

The term upland rice has nothing to do with elevation. It has become associated with sloping and hilly areas because it is usually found in the uplands. Upland is usually found in sloping land with erosion problems and drought prone. Soils are typically dry, generally acidic, fragile, and less fertile. Weeds and diseases abound. The main source of water is rain.

On the other hand, upland rice can be grown in low-lying lands that are unbunded. Compared with lowland rice, upland rice has lower yield, partly because rice as a plant depends on nutrients that are dissolved in soil moisture. Upland rice usually yield from 1 to 2 t/ha.

One of the government's missions is to help ensure the country's security in rice. Tapping the potential of the upland rice areas could help feed communities living in the rolling and mountainous terrains of the Philippines. Hence, the Upland Rice Development Program (URDP) was crafted.

The URDP aims to harness the potential of the upland rice ecosystem as one source of the country's rice supply; promote sustainable farming systems and practices in the upland communities thus increasing the farmers' income; develop the upland peoples as self-sufficient food communities; and establish a seed propagation program and protocols for traditional and modern rice varieties released for cultivation in the upland ecosystem.



KEYCHECK 1: Used high quality seeds of recommended traditional and modern upland rice varieties.

Seeds that are pure, clean, full, and uniform in size and have minimum germination rate of 85% result in less weeds, healthy seedlings, stronger to pest resistance, fast and uniform growth, and 10-15% increase in yield.





Popular traditional rice varieties

Palaweña, Asuzena, Dinorado

Popular modern rice varieties

NSIC Rc9, UPL RI 5, NSIC Rc11, Rc23, Rc25, Rc27, Rc29, Rc192

ASSESSMENT OF KEYCHECK: Used high quality seeds of traditional and modern upland rice varieties recommended in the locality or have tested and proven resistant and high yielding (at least 2- 2.5 t/ha).

Traditional upland rice varieties

Regions	Varieties
CAR	Palawan, Mimis, Azucena, Pinilisa
1	Palawan, Kamuros, Inumay
2	Palawan, Mimis, Galo, Kamuros, Pinilisa
3	Palawan, Galo, Binernal white, Dinurado, Binundok, Kalibo, Balatinaw
4A	Binerhen, Kinamuros, Kinandang, Inipot-ibon, Pirurutong
4B	Kamuros, Inipot-ibon, Inasucena, Dinurado, Milagrosa
5	Dinorado, Palawan, Gios, Binerhen
6	Dinorado, Malido, Manumbalay, Azucena, Palawan
7	Dinorado, Kamuros, Azucena,
8	Kalinayan, Baysilanon, Kanukot
9	Dinorado, Remulites, Mimis, Zambales, Azucena
10	Dinorado, Azucena, Dumudao, Palawanon, Mimis, Maria Gakit
11	Dinorado, Peria, Remulites, Azucena
12	Dinorado, Azucena, Hinomay, Kasagpi
13	Dinorado, Azucena, Mimis, Remulites
ARMM	Dinorado, Hinumay, Azucena, Kiraban, Kapukaw

Modern upland rice varieties

Varieties	Average Yield (t/ha)	Max. Yield (t/ha)	Maturity (Days after Sowing)
PSB Rc1 (Makiling)	2.4	3.9	121
PSB Rc3 (Ginilingan Puti)	2.9	6.0	123
PSB Rc5 (Arayat)	2.9	4.2	122
NSIC Rc9 (Apo)	2.9	5.5	119
NSIC Rc11 (Canlaon)	2.6	4.9	125
UPLR 1 to 7	2.8	4.6	
NSIC Rc23	2.9		108
NSIC Rc192 (Rainfed lowland)	3.7	5.5	106



KEYCHECK 2: Fields prepared on time.

Good land preparation is important for early growth of rice and promotes vigorous seedlings and uniform plant growth.

1. Plow/rotovate the land after harvest of previous crop. For flat lands, plow once, and disc-harrow twice after the first rains to make good tilth and for better crop growth.

2. Construct furrows 30-40 cm apart, depending on weed control implement available.

3. For sloping topographics, contour-plow and disc-harrow once. Construct hedge rows across the slope using strips fodder grass or perennial plants of short stature to prevent soil erosion.









KEYCHECK 3: Practiced synchronous planting with upland and lowland rice areas.

Synchronous planting within one month avoids the overlapping incidence of insect and disease populations.



ASSESSMENT OF KEYCHECK: The field had been planted 14 days before and after the majority of the upland and lowland rice areas have been planted to prevent the spread of pest damage.



KEYCHECK 4: Sufficient number of healthy seedlings.



Furrow seeding

Cover the seeds by lightly sweeping soil towards the seed to cover it.

Manual broadcasting

Cover the seeds by lightly harrowing the field after seeding.

Healthy seedlings compete better against weeds, and have better root growth, resulting in a more efficient nutrient use.



Drill seeding

Make a hole using a pointed pole with an average distance of 25-30 cm apart. Drop 5-7 seeds per hole, and lightly cover with thin soil. Land preparation is an option.





Machine seeding

Machine seeding only needs 20-60 kg of seeds per hectare. Space between seeds should be uniform to produce vigorous seedlings.

ASSESSMENT OF KEYCHECK: Direct dry seeded at 30 days after seeding achieved at least 80 healthy seedlings/m^2 for a seeding rate of 40-60 kg/ha (5-7 seeds at 25 cm x 25 cm hill distance; 65 seedlings/m^2 for a furrow seeding at 25-30 cm furrow distance).

Advantages of seeding in rows:

- Easy weeding and application of other inputs (*i.e.,* fertilizers and pesticides).
- Promotes good stand and higher yield.



KEYCHECK 5: Sufficient nutrients at tillering to early maturing, panicle initiation and flowering.

- Broadcast organic materials/fertilizer evenly after plowing.
- Apply the right amount of organic and inorganic fertilizers at appropriate growth stage.
- Fertilizer application should be based on soil analysis and or production experience.
- For organic rice cultivation of traditional rice varieties, incorporate any available animal manure (1.5 t/ha), farm biomass, biofertilizers, vermicast, green manure (legumes), trichoderma-based compost, and indigenous micro-organism.

Direct dry seeded: At ripening, panicle density should be at least 80 panicles/m^2 for traditional rice varieties and 150 panicles/m^2 for modern upland rice variety to achieve yield of 2-3 t/ha for traditional varieties and 3-4 t/ha for modern varieties.

ASSESSMENT OF KEYCHECK: Sufficient nutrient at tillering to panicle initiation and flowering.



KEYCHECK 6: No frequent and excessive rainfall or less rainfall resulting in water stress that could affect the growth and yield of the crop.



- Excessive rainfall results in less tiller, stunted growth, discolored, and unfilled-half-filled grains at ripening.
- Insufficient water causes drying and rolling of leaves.

ASSESSMENT OF KEYCHECK: No symptoms of stress due to excessive rainfall and drought stress.





KEYCHECK 7: No significant yield loss due to pests.

Integrated Pest Management Recommendations

- Use pest-resistant rice varieties.
- Use clean, purified, and healthy seeds.
- Observe field sanitation; remove alternate hosts, such as weeds.
- Monitor fields regularly for signs and symptoms of pest infestation, population of insect pests vs. natural enemies, damage and plant nutrition assessment.
- Use pesticides as a corrective measure during pest outbreak.

Know how the rice crop interacts with the biotic factors, agro-ecosystem, and crop management system.

Correctly identify pest and integrated crop management.

ASSESSMENT OF KEYCHECK: No significant yield loss due to insect pests, diseases, weeds, rats, snails, and birds. Significant pest damage occurs when one or more pests cause damage.

Metarhizium anisopliae: Microbial control agent for rice black bug



- Attacks rice black bug (RBB) with its infectious green spores that germinate once they come in contact with the RBB's skin
- Forms extensive hyphal growths within the RBB's body that penetrate into its blood vessels and invade its body cavity
- Produce toxins that paralyze and kill RBB

Advantages

- Cheaper than synthetic insecticides
- Does not leave any residue on the crops that may be harmful to humans and applicable to organic agriculture
- One preparation of *Metarhizium* contains 2.5 trillion spores. One planting season requires only two to three applications, whereas five to six applications are required for chemical pesticides.

Directions for use

- Use 4 sachets of *M. anisopliae* for every sprayer load.
- When mixing, partially fill the sprayer tank to half level and add 1 teaspoon detergent powder.
- Pour the entire content of four sachets into the tank with continuous stirring and agitation.
- Fill the tank to its required level (16 L/tankload) and continue to agitate.

Application

- Spray the solution on the rice paddy early in the morning or late in the afternoon and ensure contact between the fungus and the RBB.
- Use 40 sachets or 200 grams of *M. anisopliae* per hectare.

Metarhizium spores are like dust particles that are easily blown at the slightest movement. When adding soap or detergent solution, extra care should be given in such a way that escape of the spores could be prevented.

Disease Management


Blast


Brown spot

Disease Management Recommendations

- Diagnose disease correctly.
- Practice field sanitation.
- Only use pesticide as a corrective and not as a preventive measure.
- Monitor field from time to time.
- Observe one month fallow period to minimize disease occurrence.
- Avoid too much fertilizer, especially nitrogen.

CONSIDER

Disease distribution, spread, occurrence, condition of the field, presence of other organisms, toxic substances, and disease symptoms.

Weed Management


Umbrella sedge


Jungle rice

- Use weed-free seeds.
- Apply pre-emergence herbicide 1-2 days after seeding. Make sure the soil is moist when applying herbicide.
- Do follow-up application of post emergence herbicide when seeds are at 2-leaf stage or as needed.
- Spot weed when needed.
- Do inter-row cultivation plus hand weeding 14 and 28 days after emergence.
- Continue weeding until the rice canopy close in approximately 6 to 7 weeks after seed emergence.

Some recommended herbicides for upland rice:

- Oxadiazon (Ronstar) at 1 kg a.i/ha (4 L/ha)
- Pendimethalin (Herbadox) at 1-2 kg a.i./ha (3-6 L/ha) to control grasses particularly *R. cochinchinensis*
- 2,4-D for broadleaf and sedge control
- In areas where *Imperata sp.* (cogongrass) is a problem, application of glyphosate is recommended.

Mulching or ground covering with legume plants can play a very important role in improving the soil fertility and managing perennial weeds.

Other Pests


Bird

Bird Management

- Birds are abundant in rice fields during the ripening phase.They damage the panicles and eat the grains.They also eat germinating seeds.
- Scaring away is a practical solution.
- Practice synchronous planting.


Rat

Rat Management

- Sanitize fields, implement community rat hunting, or install a community trap barrier system (CTBS).
- Use acute rodenticide only during outbreaks.
- Plant crops, such as adlay and corn along the borders of the field, which will serve as trap crops.



KEYCHECK 8: Cut and threshed the crop at the right time.

Harvest panicles at least 85% maturity, thresh immediately, and dry at 14% moisture content for milling and 12% for planting.

EARLY HARVESTING	LATE HARVESTING
Large percent of immature and half-filled grains	Grain shattering
Low milling recovery	Milling breakage
Low grain quality	Low grain quality

Thresh the *palay* not later than 1-2 days after harvesting, especially for modern upland varieties or keep harvested panicles in bundles for sundrying and threshing.



PhilRice SACLOB, 1-ton capacity



Super Grain Bag, 40 to 60-kg capacity



Cocoon, 5-ton capacity

Hermetic Storage

- Hermetic (water and gas-tight) storage controls insects without the use of chemicals, prevents mold growth, and maintains seed germination for 6 to 8 months.
- Clean and store seeds in air-tight, cool, dry, and rat proof containers.
- Portable, needs minimal structure, and easy to install.

Notes:

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